

addition, we undertake sensitivity analysis on parameters to uncover any inherent limitations.

Step 5: Analyze the Data

We test our hypothesis to ensure the model's appropriateness and that it provides solid estimates of market impact. The experimental tests will either confirm our (model) formulation is appropriate and accurate, rule the model out, or suggest a revision or re-formulation of the model. Quite simply, does the model work? Does it require modification or should it be thrown out?

This will ensure your model offers a solid account of reality. This step also involves sensitivity analysis, evaluating errors, and performing what-if analysis surrounding extreme cases and possibilities (e.g., stress-testing the model). Here we want to learn just about everything we can—where it works well, and where its limitations may reside. We use a control data group for comparisons—that is, we perform an “out-of-sample” test utilizing a data set not included in the calibration phase. Control groups are used everywhere. In medicine—placebo drugs. Physics and engineering—controlled experiments both with and without the factors we are seeking to understand. Mathematicians and statisticians typically hold out, say one-third of the data sample to perform “out-of-sample” testing. If the model fails to predict outcomes accurately, revise the model formulation, your hypothesis, or return to the third step. You may determine in this phase that the model/hypothesis is an inappropriate solution, which is also a valuable piece of information. See *Simulating Neural Networks*, by Freeman (1994) for statistical out-of-sample testing procedures.

Step 6: Conclusion and Communication

Scientists communicate experiment results through wide ranging mediums.

Wall Street for the most, however, fails to share technological advances, particularly if a potential profit opportunity exists. The most accurate models are usually kept under lock and key to be paid for by their top investors or utilized by in-house trading groups.

Mathematical models, like I-Star, offer researchers both a workhorse model and a set of parameters to assist in the decision making process—stock selection, portfolio construction, optimization, trading algorithms, and black box modeling.